Category Theory

A category consists of

(1) objects

(2) morphisms (domain and codomain)

(3) Composition of morphisms

The axioms for a category are:

(1) Associativity:

$$\begin{tikzcd}
\cdot \arrow[r, "f"] \arrow[rr, bend right, "g\circ f"'] \arrow[rrr, bend right=60, "h\circ(g\circ f)"'] \arrow[rrr, bend right=80, "(h\circ g)\circ f"] & \cdot \arrow[r, "g"] \arrow[rr, bend right, "h\circ g"'] & \cdot \arrow[r, "h"] & \cdot
\end{tikzcd}$$

(2) identity

$$\begin{tikzcd}
W \arrow[r, "f"] & X \arrow[r, "g"] \arrow[loop above, "1_X"] & Y
\end{tikzcd}$$

① $g\circ 1_X = g$

② $1_X \circ f = f$

$$\begin{tikzcd}
W \arrow[r, "f"] \arrow[rr, bend left, "1_X\circ f = f"'] \arrow[rrr, bend left=60, "(g\circ 1_X)\circ f = g\circ f"] \arrow[rrrr, bend left=80] & X \arrow[r, "1_X"] \arrow[rr, bend left, "g\circ 1_X = g"] & X \arrow[r, "g"] & Y
\end{tikzcd}$$

$$g \circ (1_X \circ f) = g \circ f$$

Question: why 1_X is unique?

proof: assume exist $1'_X$ s.t .....

$$\begin{tikzcd} 1_X \arrow[loop above] \arrow[d, phantom, "\bullet"] \\ 1_X' \arrow[loop below] \end{tikzcd}$$

So $1_X = 1_X \circ 1_X' = 1_X'$

this concludes the proof.

Note:

$$\begin{tikzcd} X \arrow[loop above, "1_X"] \arrow[loop right, bend left, "g"] \arrow[loop right, bend left=60, "f"] \end{tikzcd}$$

any arrows in such category can be composed.

$(\hom(X,X), \circ, 1_X)$ is a monoid!

Set^2

$$\begin{tikzcd} A \arrow[r, "f"] & C \arrow[r, "f'"] & E \\ B \arrow[r, "g"'] & D \arrow[r, "g'"'] & F \end{tikzcd}$$

$$\begin{tikzcd} \langle A, B\rangle \arrow[r, "\langle f, g\rangle"] \arrow[dr] & \langle C, D\rangle \arrow[d, "\langle f', g'\rangle"] \\ & \langle E, F\rangle \end{tikzcd}$$

$$\langle f', g'\rangle \circ \langle f, g\rangle = \langle f' \circ f, g' \circ g\rangle$$

Let C, D be categories

Product Category

$$(h\circ g)\circ f = h\circ(g\circ f)$$

$$\begin{tikzcd}
a \arrow[r, "f"] \arrow[rr, bend left, "g\circ f"] \arrow[rrr, bend left=60, "h\circ g\circ f"] & b \arrow[r, "g"] \arrow[rr, bend left, "h\circ g"] & c \arrow[r, "h"] & d \\
a' \arrow[r, "f'"'] \arrow[rr, bend right, "g'\circ f'"'] \arrow[rrr, bend right=60, "h'\circ g'\circ f'"'] & b' \arrow[r, "g'"'] \arrow[rr, bend right, "h'\circ g'"'] & c' \arrow[r, "h'"'] & d'
\end{tikzcd}$$

$$(h'\circ g')\circ f' = h'\circ(g'\circ f')$$

$$\begin{aligned}
&\langle h,h'\rangle\circ(\langle g,g'\rangle\circ\langle f,f'\rangle) \\
&= \langle h,h'\rangle\circ\langle g\circ f, g'\circ f'\rangle \\
&= \langle h\circ(g\circ f), h'\circ(g'\circ f')\rangle \\
&= \langle (h\circ g)\circ f, (h'\circ g')\circ f'\rangle \\
&= \langle h\circ g, h'\circ g'\rangle\circ\langle f,f'\rangle \\
&= (\langle h,h'\rangle\circ\langle g,g'\rangle)\circ\langle f,f'\rangle
\end{aligned}$$

This concludes the proof

$\mathbf{Set}^{\to}$

$$\begin{tikzcd}
A \arrow[r, "h"] \arrow[d, "f"'] & C \arrow[d, "g"] \\
B \arrow[r, "k"'] \arrow[ur, phantom, "\ulcorner"] & D
\end{tikzcd}
\qquad
\begin{tikzcd}
f \arrow[r, "\langle h,k\rangle"] \arrow[dr, "\langle j\circ h, l\circ k\rangle"'] & g \arrow[d, "\langle j,l\rangle"] \\
& i
\end{tikzcd}$$

$$h\circ(l\circ j) = (h\circ l)\circ g$$

$$\begin{tikzcd}
a \arrow[r, "j"] \arrow[d, "i"'] \arrow[rr, bend left, "l\circ j"] \arrow[rrr, bend left=60, "h\circ l\circ j"] & b \arrow[r, "l"] \arrow[d, "q"] \arrow[rr, bend left, "h\circ l"] & c \arrow[r, "h"] \arrow[d, "r"] & d \arrow[d, "p"] \\
e \arrow[r, "k"'] \arrow[rr, bend right, "m\circ k"'] \arrow[rrr, bend right=60, "o\circ m\circ k"'] & f \arrow[r, "m"'] \arrow[rr, bend right, "o\circ m"'] & g \arrow[r, "o"'] & h
\end{tikzcd}$$

$$(o\circ m)\circ k = o\circ(m\circ k)$$

$$\begin{tikzcd}
A \arrow[r, "h"] \arrow[d, "f"'] \arrow[rr, bend left, "g\circ h"] & B \arrow[r, "g"] \arrow[d, "k"] & C \arrow[d, "l"] \\
D \arrow[r, "e"'] \arrow[rr, bend right, "j\circ e"'] & E \arrow[r, "j"'] & F
\end{tikzcd}$$

$$l\circ(g\circ h) = (j\circ e)\circ f \quad ?$$

$$\begin{array}{ccc}
\| & & \| \\
(l\circ g)\circ h & & j\circ(e\circ f) \\
\| & & \| \\
(j\circ k)\circ h & = & j\circ(k\circ h)
\end{array}$$

N, M

$D^{(0)}$ [Sym/x3]

$R_{\square}^{(1)}[\{n/a\}]$

$R_{\square}^{(0)}[\{n/a\}]$

$R_{\square}^{(1)}[\{m/x\}]$

$$[R_{\square}](c) = \left(R_{\square}^{(0)}[(1\,c)],\ \underline{\qquad}^{\uparrow}\right)$$

$$\left(\underline{\qquad}^{\uparrow},\ R_{\square}^{(0)}[[c]]\right)$$